

EXPERIMENT 18: PERFORM EDGE DETECTION USING SOBEL MATRIX ALONG Y AXIS

Aim

To detect image edges along the Y-axis using the Sobel operator in Python and OpenCV.

Algorithm

1. Import OpenCV library.
2. Read the input image.
3. Convert the image into grayscale.
4. Apply Sobel operator along the Y-axis.
5. Convert the result into an 8-bit image.
6. Display and save the output image.

Python Code

```
import cv2

# Read image
img = cv2.imread("input.jpg")

if img is None:
    print("Image not found")
    exit()

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Apply Sobel along Y-axis
sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

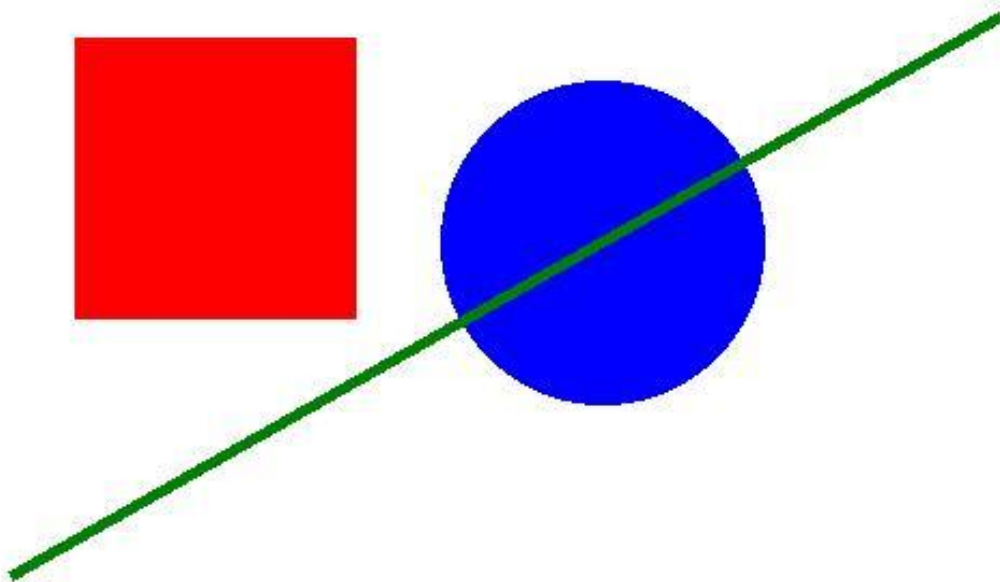
# Convert to 8-bit image
sobel_y = cv2.convertScaleAbs(sobel_y)

# Save output
cv2.imwrite("sobel_y_output.jpg", sobel_y)
```

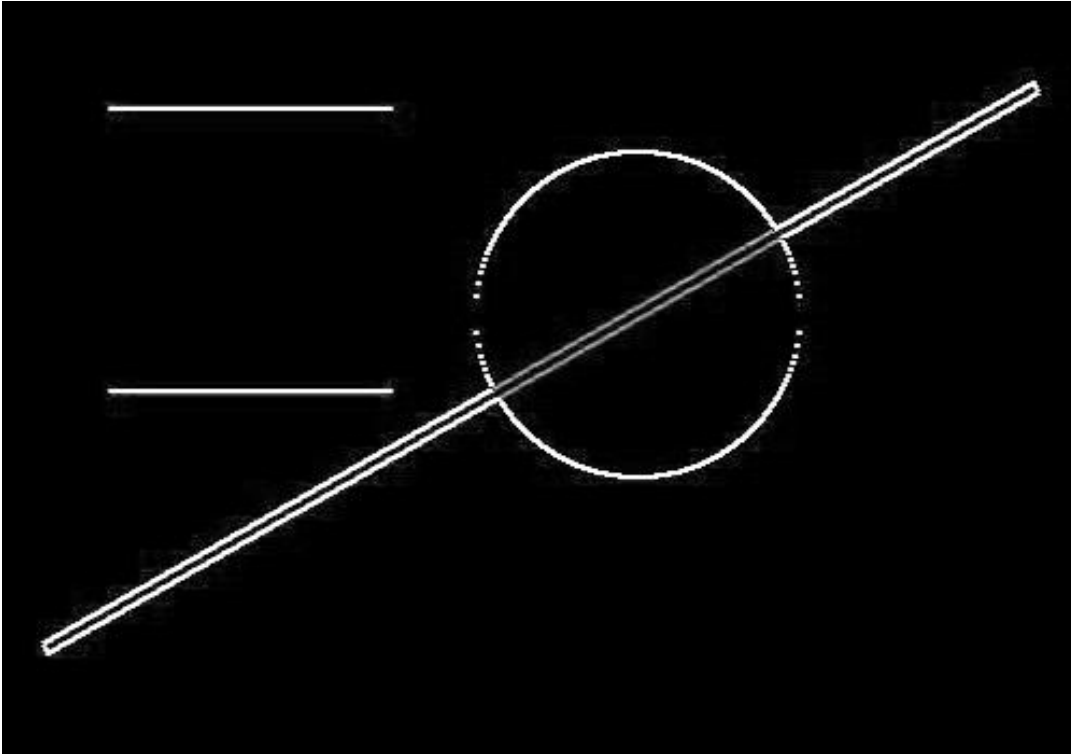
```
# Display images
cv2.imshow("Original Image", img)
cv2.imshow("Sobel Y", sobel_y)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

Input



Output



Result

Thus, the edges along the Y-axis were successfully detected using the Sobel operator.